MIL-C-10405/7A(EL)
18 September 1967

SUPERSEDING
MIL-C-10405/7(SigC)
2 April 1954

MILITARY SPECIFICATION

CRYSTAL UNIT, CR-10/U

1. SCOPE

1.1 This specification covers the detail requirements for crystal unit CR-10/U. These crystal units have a quartz vibrator pressure mounted in a metal holder Type HC-17/U, and are designed to operate at antiresonance, into a load capacitance of 25 pf, on the fundamental frequency of the vibrator, under noncontrolled temperature conditions, at the specified frequency of 5000 kilohertz(KHz).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-10405 - Crystal Units, Quartz, Pressure and Spacer Mounted.
MIL-H-10056/8 - Crystal Holder, Type HC-17/U

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

MILITARY

SC-D-17499 - Marking Instructions for Crystal Units.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5955

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3. REQUIREMENTS

3.1 Requirements shall be in accordance with MIL-C-10405 and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-C-10405.

Table 1. Detail requirements

Requirements	Specified value or condition
Holder	HC-17/U (Per MIL-H-10056/8)
Dimensions	(see MIL-H-10056/8)
Frequency range	5000.0 KHz
Frequency tolerance	0.005 percent
Mode of oscillation	Fundamental
Load capacitance	25 pf 0.5 pf
Temperature range	-40° to 70°C, incl.
Drive level, rated	20 mw 2.0 mw
Calibration Data	20ma, 50 ohms, 5000.0 kc
Equivalent resistance	90 ohms, Maximum
Government reference standard test set	TS-330/TSM
Springs	Minimum force 2-1/2, Maximum force 6 lbs.
Marking	Per SC-D-17499, method as called for in bid request and contract.
Shock:	
Frequency change permitted	0.002 percent.
	The resistance shall not exceed 90 ohms.
Vibration 1/	
Frequency change permitted	0.002 percent.
	The resistance shall not exceed 90 ohms.
Cleaning test	600 cycles (see 3.13 of MIL-C-10405)

1/ Method 201 of MIL-STD-202, amplitude 0.06 inch total excursion.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-10405.

6. NOTES

6.1 The notes specified in MIL-C-10405 are applicable to this specification.

CUSTODIANS:
ARMY - EL

PREPARING ACTIVITY
ARMY - EL
Project Nr. 5955-A048

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